Ablation : ConvNeXt-Tiny — GELU | Dropout | LayerNorm

Config: ImageNet- K pretrained | GELU activation | Dropout | LayerNorm **Model:** ConvNeXt-Ti

Dataset: Rice Leaf Disease Identification Dataset

. Colab Setup (jalankan sekali per session)

In [3]:
```
from google.colab import drive
drive.mount('/content/drive')
import os
DRIVE_DIR = '/content/drive/MyDrive/rice-convnext/03_GELU_Dropout_LayerNo
os.makedirs(DRIVE_DIR, exist_ok=True)
print(f'Drive folder ready: {DRIVE_DIR}')
```

```
Mounted at /content/drive
Drive folder ready: /content/drive/MyDrive/rice-convnext/03_GELU_Dropout_Laye
```

In [6]:
```
from google.colab import files
import os, shutil
print('Upload kaggle.json...')
files.upload()
os.makedirs('/root/.kaggle', exist_ok=True)
shutil.move('kaggle.json', '/root/.kaggle/kaggle.json')
os.chmod('/root/.kaggle/kaggle.json', 0o600)
print('Kaggle API ready!')
```

```
Upload kaggle.json...
```

Upload widget is only available when the cell has been executed in the current browser session. Pl rerun this cell to enable.

```
Saving kaggle.json to kaggle.json
Kaggle API ready!
```

In [7]:
```
DATASET_DIR = '/content/drive/MyDrive/rice-convnext/dataset'
if os.path.exists(DATASET_DIR) and len(os.listdir(DATASET_DIR)) > 0:
    print('Dataset sudah ada, skip download.')
else:
    !pip install kaggle -q
    !kaggle datasets download -d wangxiaoqii/rice-leaf-disease-identifica
    !unzip -q /content/tmp/*.zip -d {DATASET_DIR}
    !rm -rf /content/tmp
    print('Dataset ready!')

import shutil
LOCAL_DATASET = '/content/dataset'
if not os.path.exists(LOCAL_DATASET):
    print('Copying dataset ke local SSD...')
    shutil.copytree(DATASET_DIR, LOCAL_DATASET)
    print('Done!')
else:
```

```
    print('Sudah ada di local.')
DATA_DIR = LOCAL_DATASET
```

```
Dataset sudah ada, skip download.
Copying dataset ke local SSD...
Done!
```

. Setup & Imports

In [8]:
```
import os, random, shutil
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from collections import Counter
from sklearn.metrics import classification_report, confusion_matrix
import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, random_split
from torchvision import datasets, transforms, models

SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
torch.cuda.manual_seed(SEED)
torch.backends.cudnn.deterministic = True

DEVICE = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f'Device: {DEVICE}')
if torch.cuda.is_available():
    print(f'GPU: {torch.cuda.get_device_name(0)}')
```

```
Device: cuda
GPU: Tesla T4
```

. Configuration

In [9]:
```
DATA_DIR      = '/content/dataset'
DRIVE_DIR     = '/content/drive/MyDrive/rice-convnext/03_GELU_Dropout_Lay
IMG_SIZE      = 224
BATCH_SIZE    = 64
NUM_EPOCHS    = 30
LR_INIT       = 1e-3
LR_FINAL      = 1e-6
WEIGHT_DECAY  = 0.05
NUM_WORKERS   = 2
IMAGENET_MEAN = [0.485, 0.456, 0.406]
IMAGENET_STD  = [0.229, 0.224, 0.225]
SAVE_PATH     = f'{DRIVE_DIR}/convnext_tiny_03_gelu_dropout_layernorm.pt'
EXP_NAME      = 'ConvNeXt-Tiny | GELU | Dropout | LayerNorm'
print(f'Experiment: {EXP_NAME}')
```

```
Experiment: ConvNeXt-Tiny | GELU | Dropout | LayerNorm
```

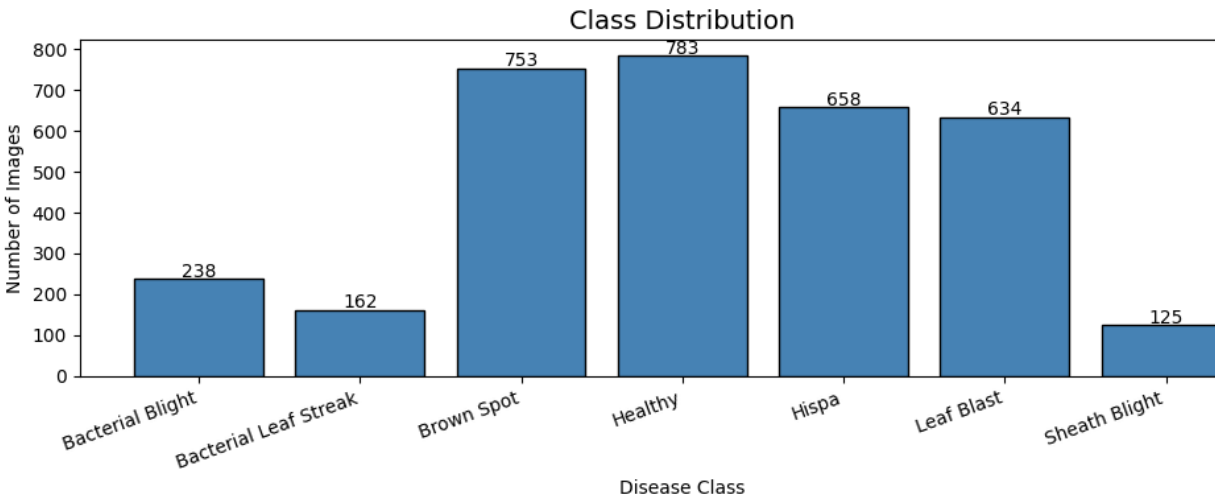
. Data Loading & Augmentation

In [10]:
```
train_transforms = transforms.Compose([
    transforms.RandomResizedCrop(IMG_SIZE),
    transforms.RandomHorizontalFlip(),
    transforms.RandomVerticalFlip(),
    transforms.RandomRotation(30),
    transforms.ColorJitter(brightness=0.2, contrast=0.2, saturation=0.2),
    transforms.ToTensor(),
    transforms.Normalize(mean=IMAGENET_MEAN, std=IMAGENET_STD),
])
val_test_transforms = transforms.Compose([
    transforms.Resize(256),
    transforms.CenterCrop(IMG_SIZE),
    transforms.ToTensor(),
    transforms.Normalize(mean=IMAGENET_MEAN, std=IMAGENET_STD),
])

full_dataset = datasets.ImageFolder(DATA_DIR)
CLASS_NAMES  = full_dataset.classes
NUM_CLASSES  = len(CLASS_NAMES)
print(f'Classes ({NUM_CLASSES}): {CLASS_NAMES}')
print(f'Total images: {len(full_dataset)}')

total   = len(full_dataset)
n_train = int(0.6 * total)
n_val   = int(0.2 * total)
n_test  = total - n_train - n_val

generator = torch.Generator().manual_seed(SEED)
train_set, val_set, test_set = random_split(
    full_dataset, [n_train, n_val, n_test], generator=generator
)

train_set.dataset.transform = train_transforms
val_set.dataset.transform   = val_test_transforms
test_set.dataset.transform  = val_test_transforms

train_loader = DataLoader(train_set, batch_size=BATCH_SIZE, shuffle=True,
val_loader   = DataLoader(val_set,   batch_size=BATCH_SIZE, shuffle=False
test_loader  = DataLoader(test_set,  batch_size=BATCH_SIZE, shuffle=False
print(f'Train: {len(train_set)} | Val: {len(val_set)} | Test: {len(test_s
```

```
Classes (7): ['Bacterial Blight', 'Bacterial Leaf Streak', 'Brown Spot', 'Hea
'Hispa', 'Leaf Blast', 'Sheath Blight']
Total images: 3353
Train: 2011 | Val: 670 | Test: 672
```

In [11]:
```
labels_all = [full_dataset.targets[i] for i in range(len(full_dataset))]
counts = [Counter(labels_all)[i] for i in range(NUM_CLASSES)]
plt.figure(figsize=(10, 4))
bars = plt.bar(CLASS_NAMES, counts, color='steelblue', edgecolor='black')
plt.title('Class Distribution', fontsize=14)
plt.xlabel('Disease Class'); plt.ylabel('Number of Images')
plt.xticks(rotation=20, ha='right')
for bar, count in zip(bars, counts):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height()+5, str(cou
plt.tight_layout()
```

```
plt.savefig('class_distribution.png', dpi=150)
plt.show()
```



. Model — ConvNeXt-Tiny (GELU | Dropout | LayerNo

In [12]:
```
from torchvision.ops import StochasticDepth

def replace_stochastic_depth(model, p=0.1):
    for name, module in model.named_children():
        if isinstance(module, StochasticDepth):
            setattr(model, name, nn.Dropout(p=p))
        else:
            replace_stochastic_depth(module, p)

model = models.convnext_tiny(weights=models.ConvNeXt_Tiny_Weights.IMAGENE
in_features = model.classifier[2].in_features
model.classifier[2] = nn.Linear(in_features, NUM_CLASSES)

# Replace Stochastic Depth → Dropout(p=0.1)
replace_stochastic_depth(model, p=0.1)
print('Regularization: StochasticDepth → Dropout(p=0.1)')

model = model.to(DEVICE)
total_p     = sum(p.numel() for p in model.parameters())
trainable_p = sum(p.numel() for p in model.parameters() if p.requires_gra
print(f'Experiment : {EXP_NAME}')
print(f'Total params    : {total_p:,}')
print(f'Trainable params: {trainable_p:,}')
```

```
Downloading: "https://download.pytorch.org/models/convnext_tiny-983f1562.pth"
root/.cache/torch/hub/checkpoints/convnext_tiny-983f1562.pth
100%|██████████| 109M/109M [00:00<00:00, 169MB/s]
Regularization: StochasticDepth → Dropout(p=0.1)
Experiment : ConvNeXt-Tiny | GELU | Dropout | LayerNorm
Total params    : 27,825,511
Trainable params: 27,825,511
```

. Training Setup

In [13]:
```
criterion = nn.CrossEntropyLoss()
optimizer = optim.AdamW(model.parameters(), lr=LR_INIT, weight_decay=WEIG
```

```
def get_scheduler(optimizer, num_epochs, lr_init, lr_final):
    warmup = optim.lr_scheduler.LinearLR(optimizer, start_factor=0.01, en
    cosine = optim.lr_scheduler.CosineAnnealingLR(optimizer, T_max=num_ep
    return optim.lr_scheduler.SequentialLR(optimizer, schedulers=[warmup,

scheduler = get_scheduler(optimizer, NUM_EPOCHS, LR_INIT, LR_FINAL)
```

. Training Loop

In [14]:

```
def train_one_epoch(model, loader, criterion, optimizer):
    model.train()
    running_loss, correct, total = 0.0, 0, 0
    for images, labels in loader:
        images, labels = images.to(DEVICE), labels.to(DEVICE)
        optimizer.zero_grad()
        outputs = model(images)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()
        running_loss += loss.item() * images.size(0)
        _, predicted = outputs.max(1)
        correct += predicted.eq(labels).sum().item()
        total   += labels.size(0)
    return running_loss / total, correct / total

def evaluate(model, loader, criterion):
    model.eval()
    running_loss, correct, total = 0.0, 0, 0
    with torch.no_grad():
        for images, labels in loader:
            images, labels = images.to(DEVICE), labels.to(DEVICE)
            outputs = model(images)
            loss = criterion(outputs, labels)
            running_loss += loss.item() * images.size(0)
            _, predicted = outputs.max(1)
            correct += predicted.eq(labels).sum().item()
            total   += labels.size(0)
    return running_loss / total, correct / total

history = {'train_loss': [], 'train_acc': [], 'val_loss': [], 'val_acc':
best_val_acc = 0.0

for epoch in range(NUM_EPOCHS):
    train_loss, train_acc = train_one_epoch(model, train_loader, criterio
    val_loss,   val_acc   = evaluate(model, val_loader, criterion)
    scheduler.step()
    history['train_loss'].append(train_loss)
    history['train_acc'].append(train_acc)
    history['val_loss'].append(val_loss)
    history['val_acc'].append(val_acc)
    if val_acc > best_val_acc:
        best_val_acc = val_acc
        torch.save(model.state_dict(), SAVE_PATH)
    print(f'Epoch [{epoch+1:02d}/{NUM_EPOCHS}] '
          f'Train Loss: {train_loss:.4f} Acc: {train_acc:.4f} | '
          f'Val Loss: {val_loss:.4f} Acc: {val_acc:.4f}'
          + (' <- best' if val_acc == best_val_acc else ''))
```

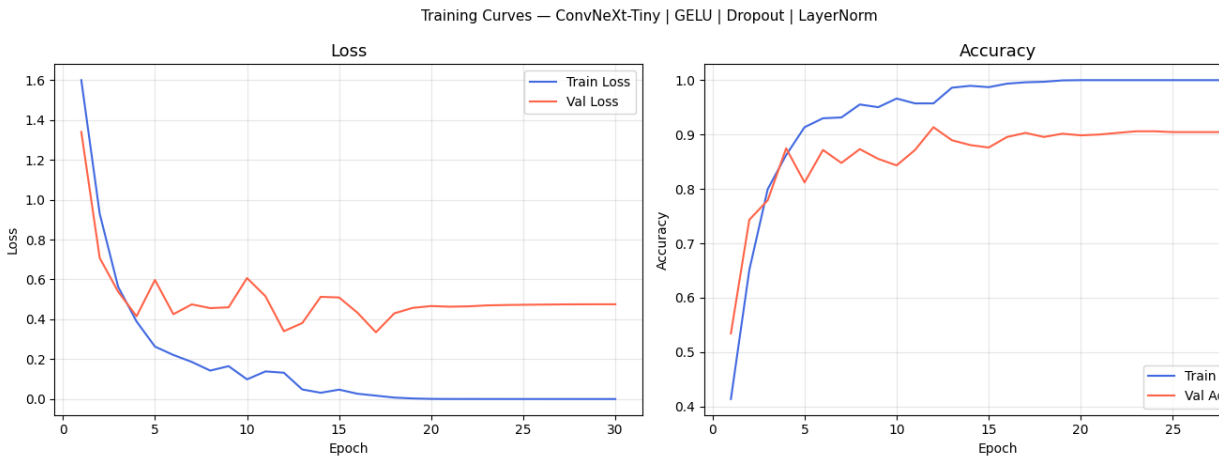
```
print(f'\nBest Val Accuracy: {best_val_acc:.4f}')
print(f'Saved: {SAVE_PATH}')
```

```
Epoch [01/30] Train Loss: 1.6000 Acc: 0.4137 | Val Loss: 1.3403 Acc: 0.5343 <
Epoch [02/30] Train Loss: 0.9297 Acc: 0.6509 | Val Loss: 0.7078 Acc: 0.7433 <
Epoch [03/30] Train Loss: 0.5634 Acc: 0.7996 | Val Loss: 0.5390 Acc: 0.7791 <
Epoch [04/30] Train Loss: 0.3893 Acc: 0.8623 | Val Loss: 0.4154 Acc: 0.8746 <
Epoch [05/30] Train Loss: 0.2632 Acc: 0.9135 | Val Loss: 0.5976 Acc: 0.8119
Epoch [06/30] Train Loss: 0.2212 Acc: 0.9299 | Val Loss: 0.4257 Acc: 0.8716
Epoch [07/30] Train Loss: 0.1862 Acc: 0.9314 | Val Loss: 0.4750 Acc: 0.8478
Epoch [08/30] Train Loss: 0.1428 Acc: 0.9552 | Val Loss: 0.4562 Acc: 0.8731
Epoch [09/30] Train Loss: 0.1650 Acc: 0.9503 | Val Loss: 0.4604 Acc: 0.8552
Epoch [10/30] Train Loss: 0.0988 Acc: 0.9662 | Val Loss: 0.6071 Acc: 0.8433
Epoch [11/30] Train Loss: 0.1386 Acc: 0.9572 | Val Loss: 0.5159 Acc: 0.8716
Epoch [12/30] Train Loss: 0.1321 Acc: 0.9572 | Val Loss: 0.3402 Acc: 0.9134 <
Epoch [13/30] Train Loss: 0.0481 Acc: 0.9861 | Val Loss: 0.3814 Acc: 0.8896
Epoch [14/30] Train Loss: 0.0316 Acc: 0.9896 | Val Loss: 0.5128 Acc: 0.8806
Epoch [15/30] Train Loss: 0.0472 Acc: 0.9871 | Val Loss: 0.5096 Acc: 0.8761
Epoch [16/30] Train Loss: 0.0267 Acc: 0.9935 | Val Loss: 0.4330 Acc: 0.8955
Epoch [17/30] Train Loss: 0.0171 Acc: 0.9960 | Val Loss: 0.3349 Acc: 0.9030
Epoch [18/30] Train Loss: 0.0073 Acc: 0.9970 | Val Loss: 0.4301 Acc: 0.8955
Epoch [19/30] Train Loss: 0.0030 Acc: 0.9995 | Val Loss: 0.4575 Acc: 0.9015
Epoch [20/30] Train Loss: 0.0009 Acc: 1.0000 | Val Loss: 0.4666 Acc: 0.8985
Epoch [21/30] Train Loss: 0.0004 Acc: 1.0000 | Val Loss: 0.4634 Acc: 0.9000
Epoch [22/30] Train Loss: 0.0004 Acc: 1.0000 | Val Loss: 0.4650 Acc: 0.9030
Epoch [23/30] Train Loss: 0.0004 Acc: 1.0000 | Val Loss: 0.4699 Acc: 0.9060
Epoch [24/30] Train Loss: 0.0003 Acc: 1.0000 | Val Loss: 0.4717 Acc: 0.9060
Epoch [25/30] Train Loss: 0.0003 Acc: 1.0000 | Val Loss: 0.4730 Acc: 0.9045
Epoch [26/30] Train Loss: 0.0003 Acc: 1.0000 | Val Loss: 0.4740 Acc: 0.9045
Epoch [27/30] Train Loss: 0.0003 Acc: 1.0000 | Val Loss: 0.4748 Acc: 0.9045
Epoch [28/30] Train Loss: 0.0003 Acc: 1.0000 | Val Loss: 0.4753 Acc: 0.9045
Epoch [29/30] Train Loss: 0.0003 Acc: 1.0000 | Val Loss: 0.4755 Acc: 0.9045
Epoch [30/30] Train Loss: 0.0003 Acc: 1.0000 | Val Loss: 0.4755 Acc: 0.9045

Best Val Accuracy: 0.9134
Saved: /content/drive/MyDrive/rice-convnext/03_GELU_Dropout_LayerNorm/
convnext_tiny_03_gelu_dropout_layernorm.pt
```

. Training Curves

In [15]:

```
epochs_range = range(1, NUM_EPOCHS + 1)
fig, axes = plt.subplots(1, 2, figsize=(14, 5))
axes[0].plot(epochs_range, history['train_loss'], label='Train Loss', col
axes[0].plot(epochs_range, history['val_loss'],   label='Val Loss',   col
axes[0].set_title('Loss', fontsize=13)
axes[0].set_xlabel('Epoch'); axes[0].set_ylabel('Loss')
axes[0].legend(); axes[0].grid(alpha=0.3)
axes[1].plot(epochs_range, history['train_acc'], label='Train Accuracy',
axes[1].plot(epochs_range, history['val_acc'],   label='Val Accuracy',
axes[1].set_title('Accuracy', fontsize=13)
axes[1].set_xlabel('Epoch'); axes[1].set_ylabel('Accuracy')
axes[1].legend(); axes[1].grid(alpha=0.3)
plt.suptitle(f'Training Curves — {EXP_NAME}', fontsize=11)
plt.tight_layout()
plt.savefig('training_curves.png', dpi=150)
plt.show()
```



. Evaluation on Test Set

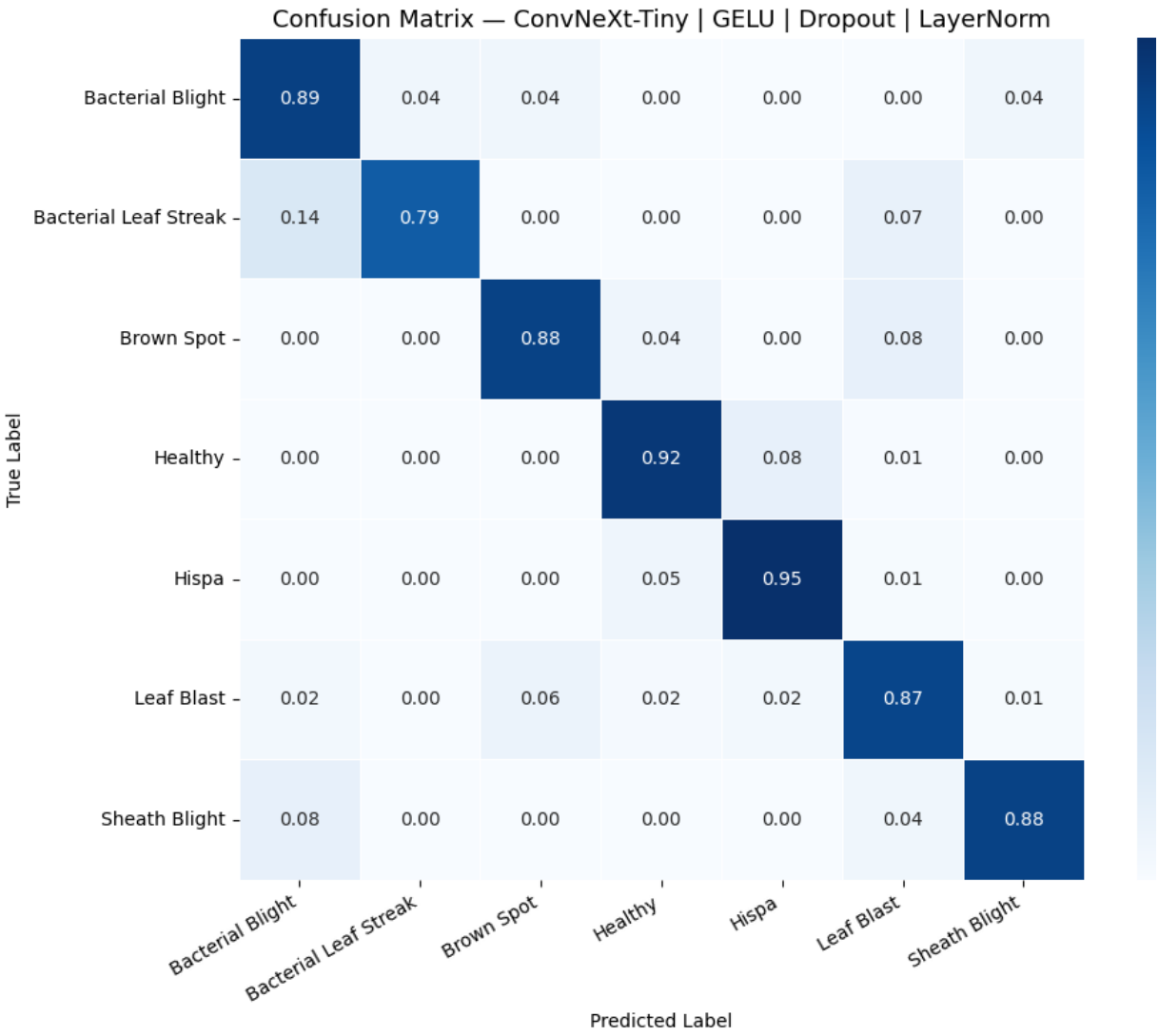
In [16]:
```
model.load_state_dict(torch.load(SAVE_PATH, map_location=DEVICE))
model.eval()
all_preds, all_labels = [], []
with torch.no_grad():
    for images, labels in test_loader:
        images = images.to(DEVICE)
        outputs = model(images)
        _, predicted = outputs.max(1)
        all_preds.extend(predicted.cpu().numpy())
        all_labels.extend(labels.numpy())

report = classification_report(all_labels, all_preds, target_names=CLASS_
report_df = pd.DataFrame(report).transpose()
print('=== Classification Report ===')
print(report_df.round(4).to_string())
report_df.to_csv('classification_report.csv')

summary = {
    'Model'     : ['ConvNeXt-Tiny'],
    'Activation': ['GELU'],
    'Dropout'   : ['Dropout'],
    'Norm'      : ['LayerNorm'],
    'Accuracy'  : [report['accuracy']],
    'Precision' : [report['weighted avg']['precision']],
    'Recall'    : [report['weighted avg']['recall']],
    'F1-Score'  : [report['weighted avg']['f1-score']],
}
summary_df = pd.DataFrame(summary)
print('\n=== Summary ===')
print(summary_df.round(4).to_string(index=False))
summary_df.to_csv('summary.csv', index=False)
```

```
=== Classification Report ===
                      precision  recall  f1-score   support
Bacterial Blight         0.8421  0.8889    0.8649   54.0000
Bacterial Leaf Streak    0.9167  0.7857    0.8462   28.0000
Brown Spot               0.9419  0.8795    0.9097  166.0000
Healthy                  0.9157  0.9157    0.9157  166.0000
Hispa                    0.8667  0.9455    0.9043  110.0000
Leaf Blast               0.8560  0.8699    0.8629  123.0000
Sheath Blight            0.8800  0.8800    0.8800   25.0000
accuracy                 0.8943  0.8943    0.8943    0.8943
macro avg                0.8884  0.8807    0.8834  672.0000
weighted avg             0.8960  0.8943    0.8944  672.0000

=== Summary ===
        Model Activation Dropout      Norm  Accuracy  Precision  Recall  F1-S
ConvNeXt-Tiny       GELU Dropout LayerNorm    0.8943      0.896  0.8943    0.
```

. Confusion Matrix

In [17]:

```python
cm = confusion_matrix(all_labels, all_preds)
cm_norm = cm.astype('float') / cm.sum(axis=1)[:, np.newaxis]
plt.figure(figsize=(10, 8))
sns.heatmap(cm_norm, annot=True, fmt='.2f',
            xticklabels=CLASS_NAMES, yticklabels=CLASS_NAMES,
            cmap='Blues', linewidths=0.5)
plt.title(f'Confusion Matrix — ConvNeXt-Tiny | GELU | Dropout | LayerNorm
plt.ylabel('True Label'); plt.xlabel('Predicted Label')
plt.xticks(rotation=30, ha='right')
plt.tight_layout()
plt.savefig('confusion_matrix.png', dpi=150)
plt.show()
```



. Save All to Drive

In [18]:
```
local_files = ['training_curves.png', 'confusion_matrix.png',
               'classification_report.csv', 'summary.csv', 'class_distrib
for fname in local_files:
    if os.path.exists(fname):
        shutil.copy(fname, f'{DRIVE_DIR}/{fname}')

all_outputs = [
    (f'{DRIVE_DIR}/convnext_tiny_03_gelu_dropout_layernorm.pt',
    (f'{DRIVE_DIR}/training_curves.png',              'Training curves'),
    (f'{DRIVE_DIR}/confusion_matrix.png',             'Confusion matrix')
    (f'{DRIVE_DIR}/classification_report.csv',        'Per-class metrics'
    (f'{DRIVE_DIR}/summary.csv',                      'Summary table'),
]
print(f'=== Output Files: {DRIVE_DIR} ===')
for fpath, desc in all_outputs:
    exists = '✅' if os.path.exists(fpath) else '❌'
    print(f'{exists}  {os.path.basename(fpath):<45} {desc}')
```

```
=== Output Files: /content/drive/MyDrive/rice-convnext/03_GELU_Dropout_LayerN
===
✅  convnext_tiny_03_gelu_dropout_layernorm.pt    Model weights
✅  training_curves.png                           Training curves
✅  confusion_matrix.png                          Confusion matrix
✅  classification_report.csv                     Per-class metrics
✅  summary.csv                                   Summary table
```